Questions

Q1.

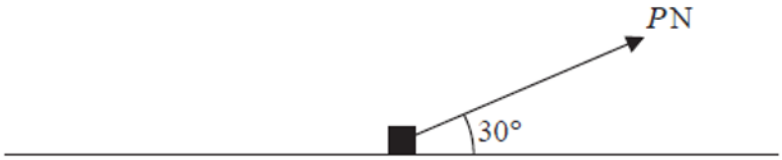


Figure 1

RocketRevise

A small block of mass 5 kg lies at rest on a rough horizontal plane.

The coefficient of friction between the block and the plane is $\frac{3}{7}$

A force of magnitude P newtons is applied to the block in a direction which makes an angle of 30° with the plane, as shown in Figure 1.

The block is modelled as a particle.

Given that $P = 14$

(a) find the magnitude of the frictional force exerted on the block by the plane and describe what happens to the block, justifying your answer.

(6)

The value of P is now changed so that the block is on the point of slipping along the plane.

(b) Find the value of P

(6)

(Total for question = 12 marks)

www.RocketRevise.com

(Q04 WME01/01, June 2022)

Q2.

compiled by Ahmed Gul
www.RocketRevise.com

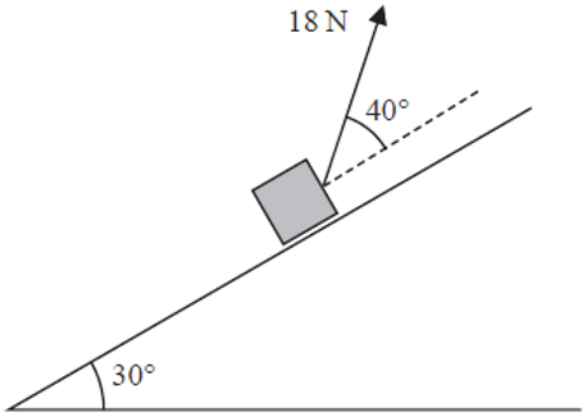


Figure 5

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

RocketRevise

The force has magnitude 18 N and acts at an angle of 40° to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle P

(a) Find the acceleration of P

(8)

The points A and B lie on a line of greatest slope of the plane, where $AB = 5$ m and B is above A. Particle P passes through A with speed 2 m s^{-1} in the direction AB.

(b) Find the speed of P as it passes through B.

(3)

The force of 18 N is removed at the instant P passes through B. As a result, P comes to rest at the point C.

(c) Determine whether P will remain at rest at C. You must show all stages of your working clearly.

www.RocketRevise.com

(4)

(Total for question = 15 marks)

(Q08 WME01/01, Jan 2023)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q3.

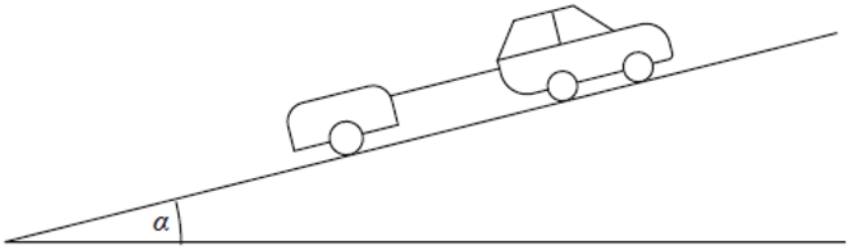


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin\alpha = \frac{1}{12}$

RocketRevise

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration 0.75 m s^{-2}

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude 2*R* newtons
- the **trailer** is modelled as a constant force of magnitude *R* newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of *R* is 60

(4)

(b) find the tension in the tow bar.

www.RocketRevise.com

(3)

When the car and trailer are moving at a speed of 12 m s^{-1} , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

(Total for question = 11 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(**Q07 WME01/01, June 2023**)

Q4.

A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan\theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 ms $^{-1}$ from a point A on the plane, up a line of greatest slope of the plane.

RocketRevise

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

(a) Find the magnitude of the frictional force acting on the particle as it moves up the plane.

(3)

The particle comes to instantaneous rest at the point B.

(b) Find the distance AB.

(5)

The particle now slides down the plane from B. At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 ms $^{-1}$

(c) Find the distance BC.

(5)

(Total for question = 13 marks)

www.RocketRevise.com

(**Q06 WME01/01, June 2021**)

Q5.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

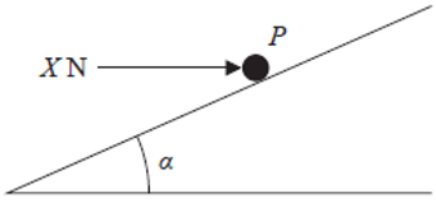


Figure 2

A rough plane is inclined to the horizontal at an angle α, where tan α = 3/4

A particle *P* of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude *X* newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

(a) Show that when $X = 14.7$ there is no frictional force acting on *P*

RocketRevise

(3)

The coefficient of friction between *P* and the plane is 0.5

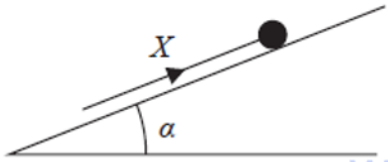
(b) Find the smallest possible value of *X*.

(8)

(Total for question = 11 marks)

(Q03 WME01/01, Oct 2022)

Q6.



www.RocketRevise.com

Figure 3

A particle of mass *m* rests in equilibrium on a fixed rough plane under the action of a force of magnitude *X*. The force acts up a line of greatest slope of the plane, as shown in Figure 3.

The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$

The coefficient of friction between the particle and the plane is μ.

- When $X = 2P$, the particle is on the point of sliding up the plane.

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

- When $X = P$, the particle is on the point of sliding down the plane.

Find the value of μ.

(Total for question = 8 marks)

(Q05 WME01/01, Jan 2022)

RocketRevise

www.RocketRevise.com

www.RocketRevise.com